Townsend 4.4-4.7

Questions for Gallicchio

I don't understand how the basis for the magnetic field Hamiltonian was constructed, nor how we made the ammonium up vs down states out of states II and I.

4.4

Spin $\frac{1}{2}$ particles will precess in a magnetic field like a gyroscope.

The spin Hamiltonian for magnetic resonance is given by

$$\hat{H} = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} = -\frac{gq}{2mc}\hat{\mathbf{S}} \cdot \mathbf{B} = -\frac{gq}{2mc}\hat{\mathbf{S}} \cdot (B_1 \cos \omega t \; \mathbf{i} + B_0 \mathbf{k}) \qquad (4.34)$$

We have a time dependent equation that shows us the precession of the electron in a magtnetic field.

4.5

We can make a spin 1 particle out of NH3 - the N can be above or below a plane with equal probability. We can find eigenstates and build the representation for the above vs below states.
In a time evolving field, stuff happens

4.6

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

where Δt is the minimum time for the state to evolve. For an energy eigenstate $\Delta = 0$ so $\Delta t = \infty$ so it never evolves.

4.6